

Simone Carletti

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About Me

I am a M.Sc. student in **Computer Engineering** specializing in **Robotics and Autonomous Systems**. I am currently completing my master's thesis at EPFL, developing an adaptive flight control system for a **morphing-wing drone**, giving me hands-on familiarity with Reinforcement Learning, Control and UAV dynamics. I have a long-standing passion for **hardware**, having worked extensively with embedded boards like Arduino, RPi, ESP8266, etc. Beyond my academic work, my **professional experience** as a lead developer has allowed me to build strong software engineering skills, focusing on clean and maintainable code. I am proficient in **C++** and **Python**, with extensive experience with **Linux** environments, utilizing **Git**, and **debugging** complex multi-threaded systems.

Education

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| EPFL - LIS , <i>Master's Thesis</i> | <i>Sept 2025 – Present</i> |
| <ul style="list-style-type: none">◦ Title: $\mathcal{L}_1$-Adaptive augmentation for robust morphing-wing drone flight control◦ Skills: Control Systems, RL (PPO), C++, Python, Simulink, ROS2, Git, Linux, Unity | |
| DTU , <i>M.Sc. Autonomous Systems</i> | <i>Sept 2024 – Jul 2025</i> |
| <ul style="list-style-type: none">◦ Erasmus+ exchange - Converted GPA: 30/30 | |
| Politecnico di Torino , <i>M.Sc. Computer Engineering</i> | <i>Sept 2023 – Present</i> |
| <ul style="list-style-type: none">◦ Curriculum in Automation and Intelligent Cyber-Physical Systems - Final GPA: 29.48/30 | |
| Università degli Studi di Firenze , <i>B.Sc. Computer Science and Engineering</i> | <i>Sept 2020 – Sept 2023</i> |
| <ul style="list-style-type: none">◦ Graduated with 110/110 cum laude (with honors) | |

Selected Projects

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| Multiplayer videogame on embedded hardware (GitHub 🔗) | <i>Jan 2024 – Jun 2024</i> |
| <ul style="list-style-type: none">◦ Developed bare-metal C firmware on ARM Cortex-M3 (LandTiger) using Keil μVision◦ Designed and implemented a custom real-time communication protocol over CAN bus to enable multiplayer support on separate boards.◦ Ensured deterministic behavior under strict timing and memory constraints. | |
| Digital Control - Autonomous Robotics (GitHub 🔗) | <i>Oct 2024 – Dec 2024</i> |
| <ul style="list-style-type: none">◦ Designed, simulated, and tuned digital control systems using MATLAB and Simulink to ensure robust mathematical performance before hardware deployment.◦ Implemented the finalized control algorithms in C/C++ on an embedded microcontroller, successfully bridging the gap between theoretical simulation and real-time execution. | |
| Localization with EKF in ROS2 | <i>Nov 2023 – Dec 2023</i> |
| <ul style="list-style-type: none">◦ Implemented a full EKF localization pipeline from scratch in ROS2 and Python, fusing data from LiDAR, IMU, and wheel odometry | |
| Autonomous Navigation & Task Execution - DTU RoboCup 🔗 | <i>Feb 2025 – May 2025</i> |
| <ul style="list-style-type: none">◦ Developed autonomous navigation, control, and task execution algorithms with continuous integration and validation tests on the real platform | |

Experience

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|---|--|
| Robotics Engineer , <i>DIANA student team</i> | <i>Sept 2023 – Jul 2024</i>
<i>Turin, Italy</i> |
| <ul style="list-style-type: none">◦ Collaborated on a Mars rover student project, implementing perception and EKF state estimation in C++ while contributing to software development, testing, and system integration | |
| Full-Stack Developer , <i>Genomedics S.r.l</i> | <i>Sept 2020 – Present</i>
<i>Florence, Italy</i> |
| <ul style="list-style-type: none">◦ Lead developer of three key projects and for the company's main project (KPlatform)◦ Skills: Angular (Typescript, SCSS, HTML), Vue.js, PHP, Laravel, SQL, Python, Git, Linux | |

Skills & Activities

- **Languages:** Fluent in English, Native Italian speaker, Basic Spanish skills
- **Programming:** C++, C, Python, TypeScript, JavaScript, Rust, Java, Matlab, PHP, SQL, HTML, CSS/SCSS, Assembly, LaTeX
- **Tools:** PyTorch, Linux, Git, Google Colab, OpenCV, Onshape, CMake, Docker, Node-RED, Multisim
- **Other:** Raspberry Pi, Arduino, ESP8266, NodeMCU, Soldering, Nvidia Jetson/Orin, Guitar, Photography, Networking, Zigbee